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Chapter 1

Manufacturing Overview



Introduction

The JBA Production system covers the complete spectrum of enterprise management and comprises four key business functions:

- Finance.
- Logistics.
- Production.
- Customer Service.

These functions are incorporated within a business cycle of activities illustrated in Figure 1. The array of associated functional applications may be used to support management of activities across the generic supply chain as shown in Figure 2. Integration of data enabling the generation of a comprehensive range of printed business reports taking differing perspectives of the enterprise is fundamental to supply chain management, and this is effectively achieved within the system. A simplified view of the organisation and inter-relationships of a fully integrated Production driven business system which makes management data available is shown in Figure 3.

Figure 1 - Activity Cycle

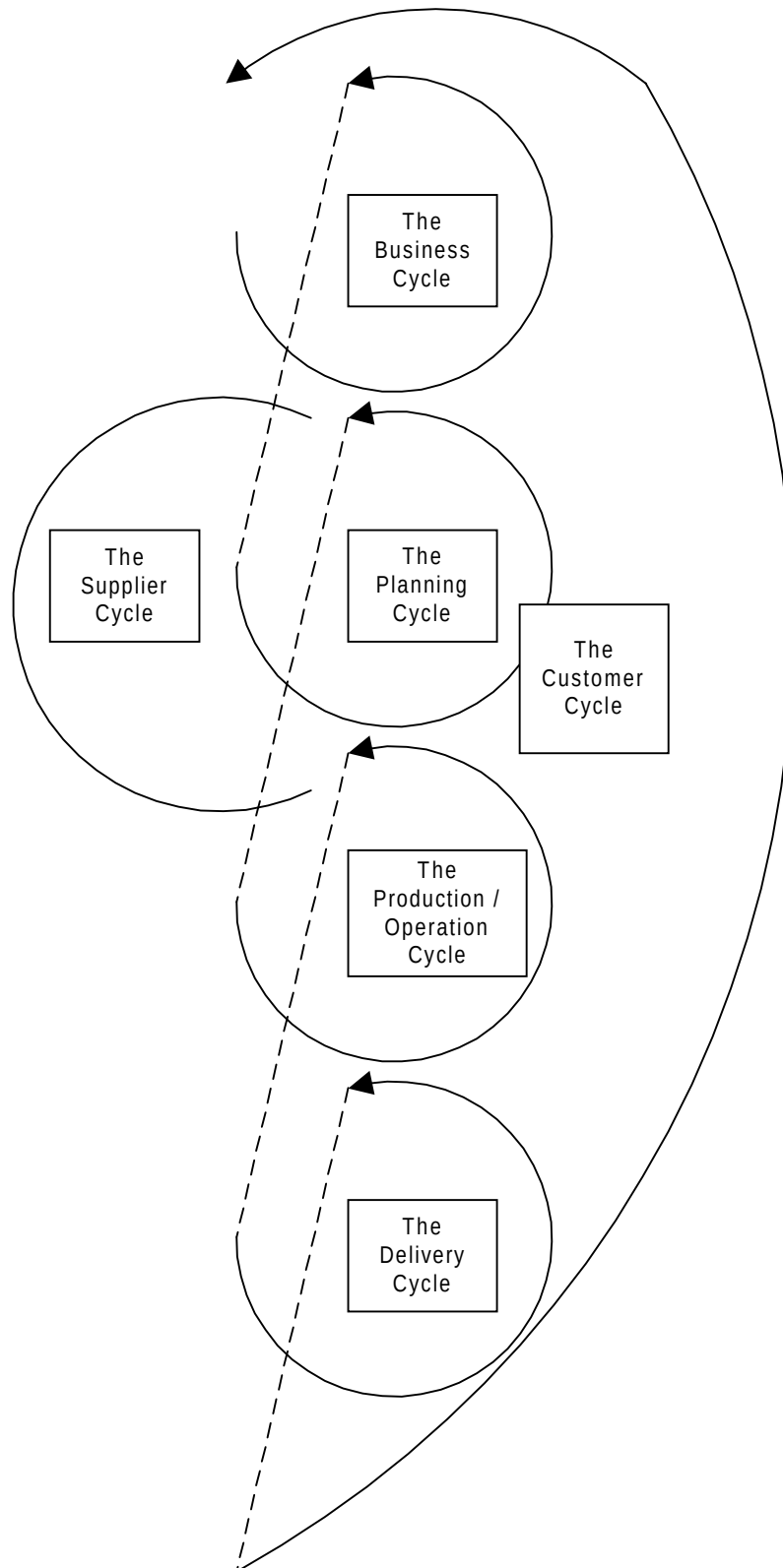
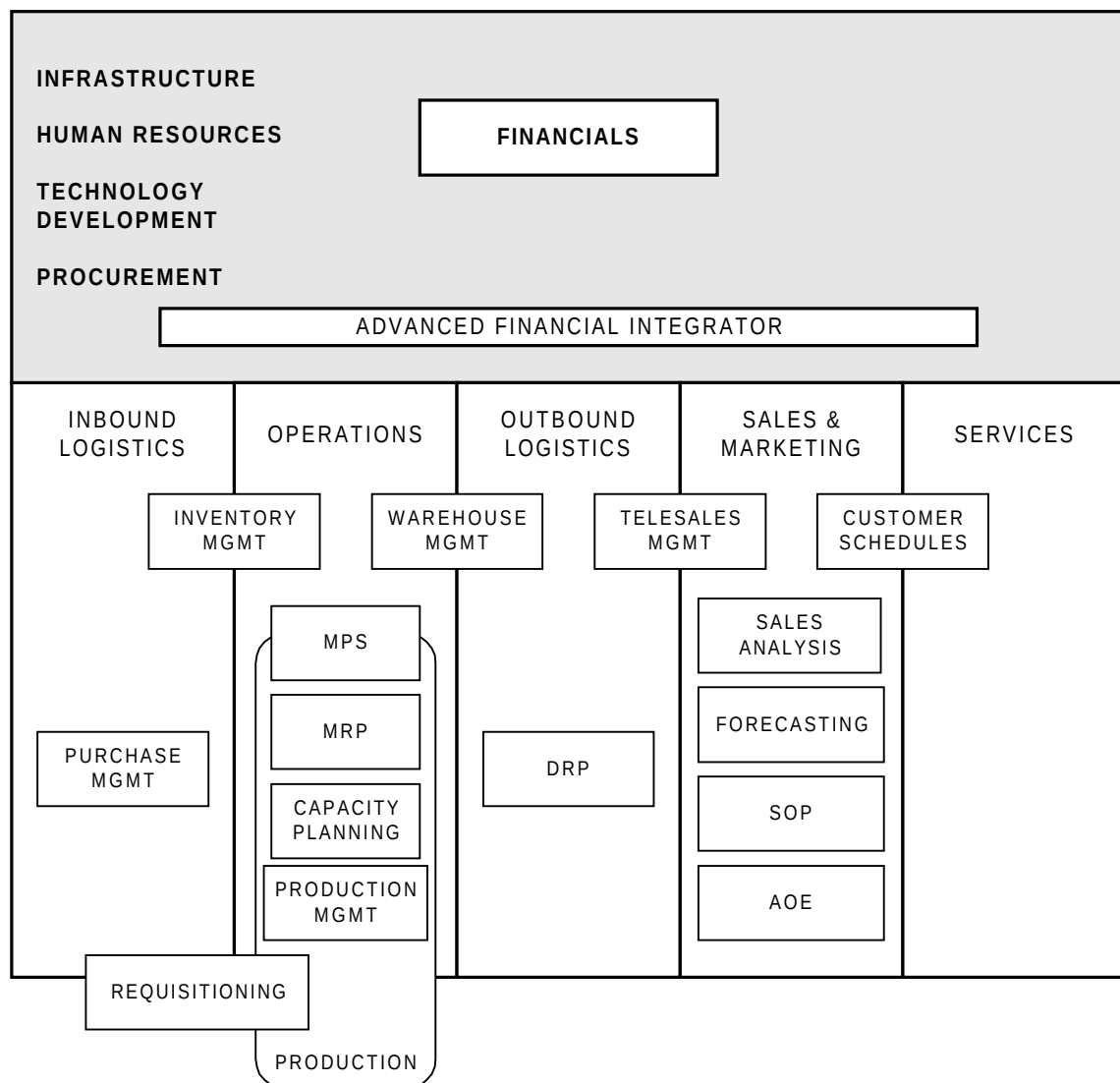
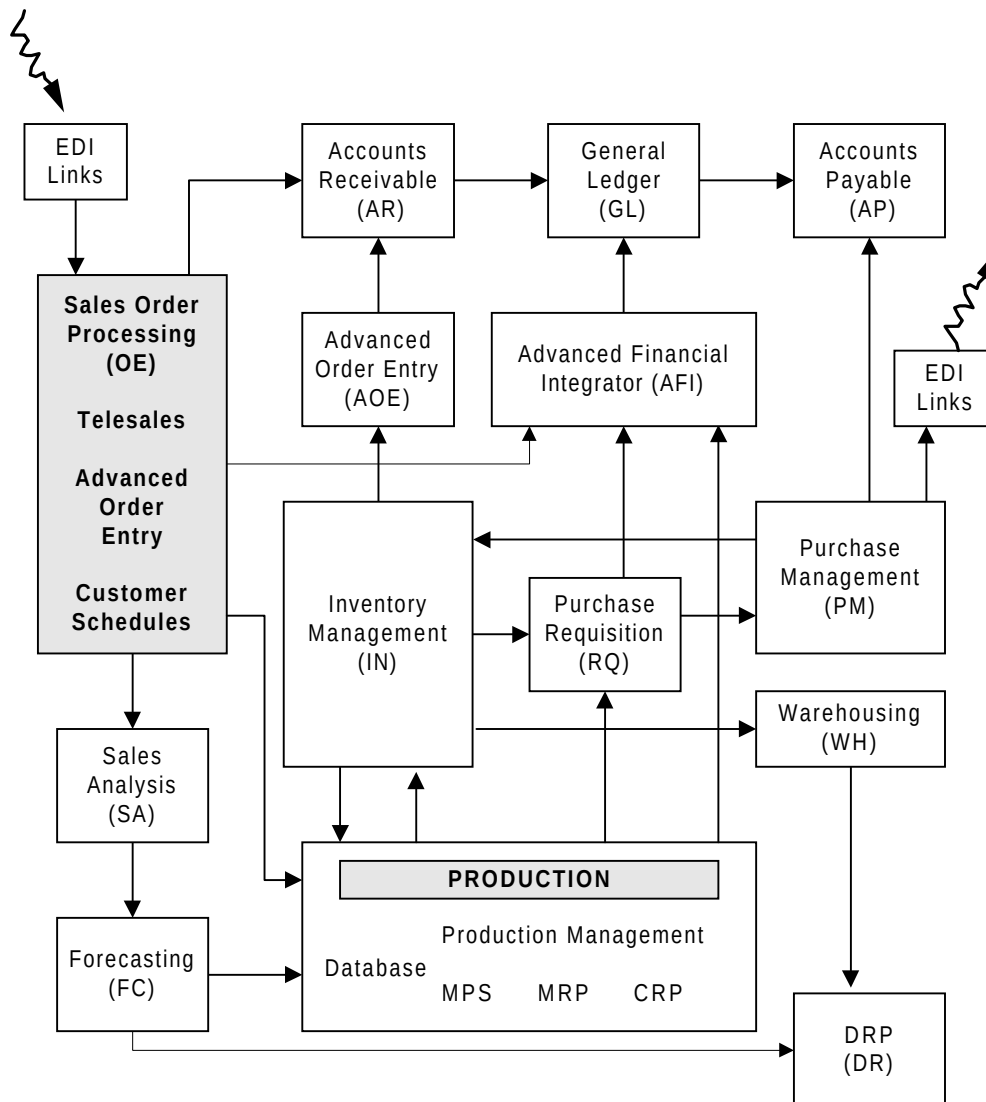


Figure 2 - JBA Applications & the Generic Supply Chain



- MPS Master Production Scheduling
MRP Material Requirements Planning
SOP Sales Order Processing
AOE Advanced Order Entry
DRP Distribution Requirements Planning

Figure 3 - A Fully Integrated Production System



Production System Applications

The Production system, Figure 4, comprises six core applications, governed by a Company Profile. The link between Production applications and Inventory applications is mandatory, whilst the others shown are optional, but functionally desirable, for the effective integration of management information.

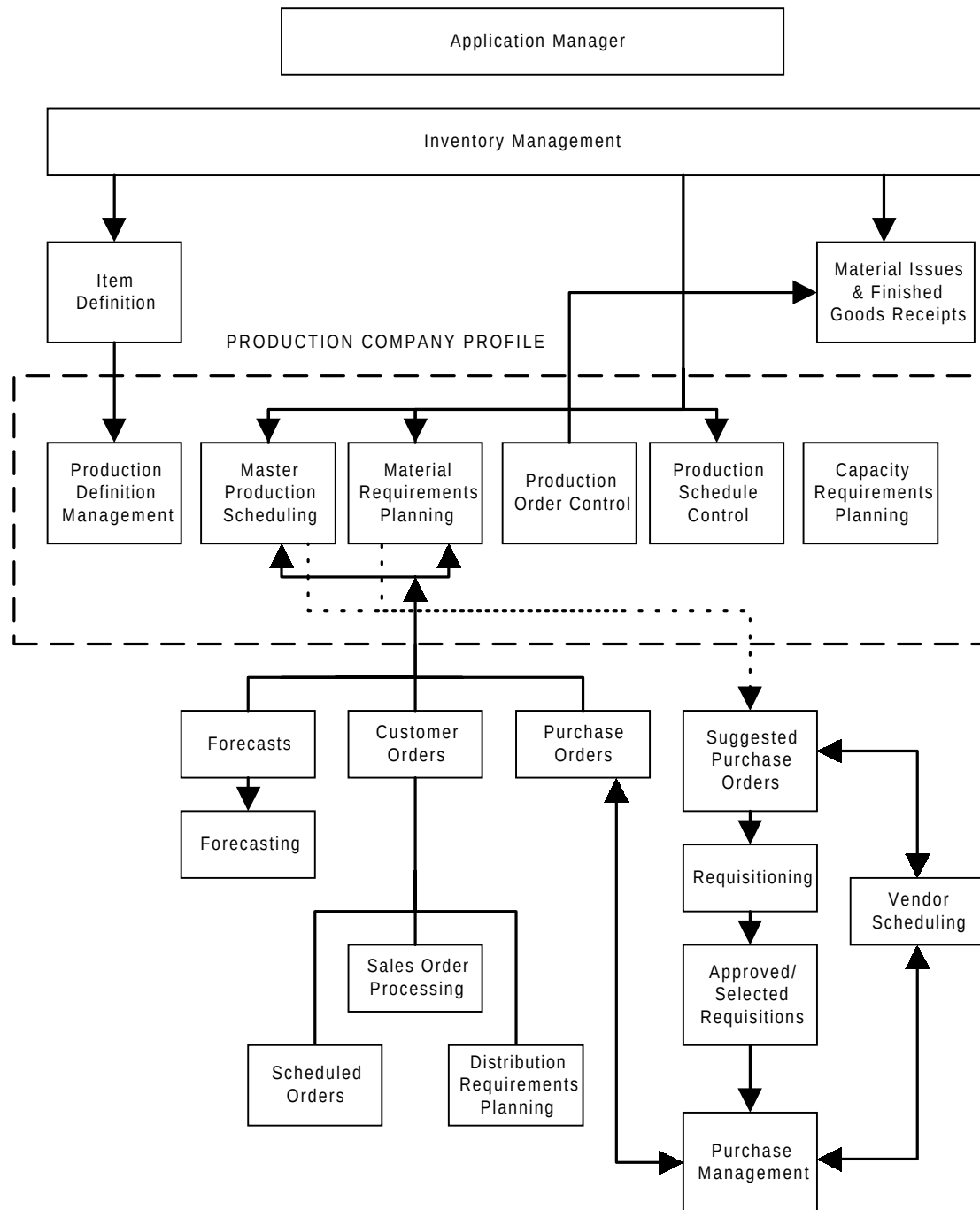
Company Profile

The Production system, in common with all JBA applications, operates within a multi-company environment.

A Company Profile sets the basic system defaults and operation policies relevant to each Production company; defines the costing elements, and defines base calculation parameters which are used by all Production functions.

Access to the Company Profile is usually restricted to system managers and implementation project managers who require a comprehensive understanding of the implications of each of its parameter settings. However, awareness of the Company Profile's purpose and features is relevant to all users.

Figure 4 - The Production System



Operation Management Cycle

The Operations Management cycle addressed by the Production system is summarised in Figure 5. The diagram identifies the two methods of production control available: Production Order Control and Production Schedule Control. In general, the former is more appropriate to flexible, low to medium volume product structures, and the latter, to repetitive, medium to high volume product structures; both of which may be found in a single organisation and both of which may be catered for, running concurrently within the Production system.

Production Order Control

Sales orders/customer schedules and forecasts are combined to create a demand input to the Master Production Scheduling (MPS) application. A high level plan is created which suggests production and purchase order quantities and dates. This plan normally consists of only customer deliverable or saleable items.

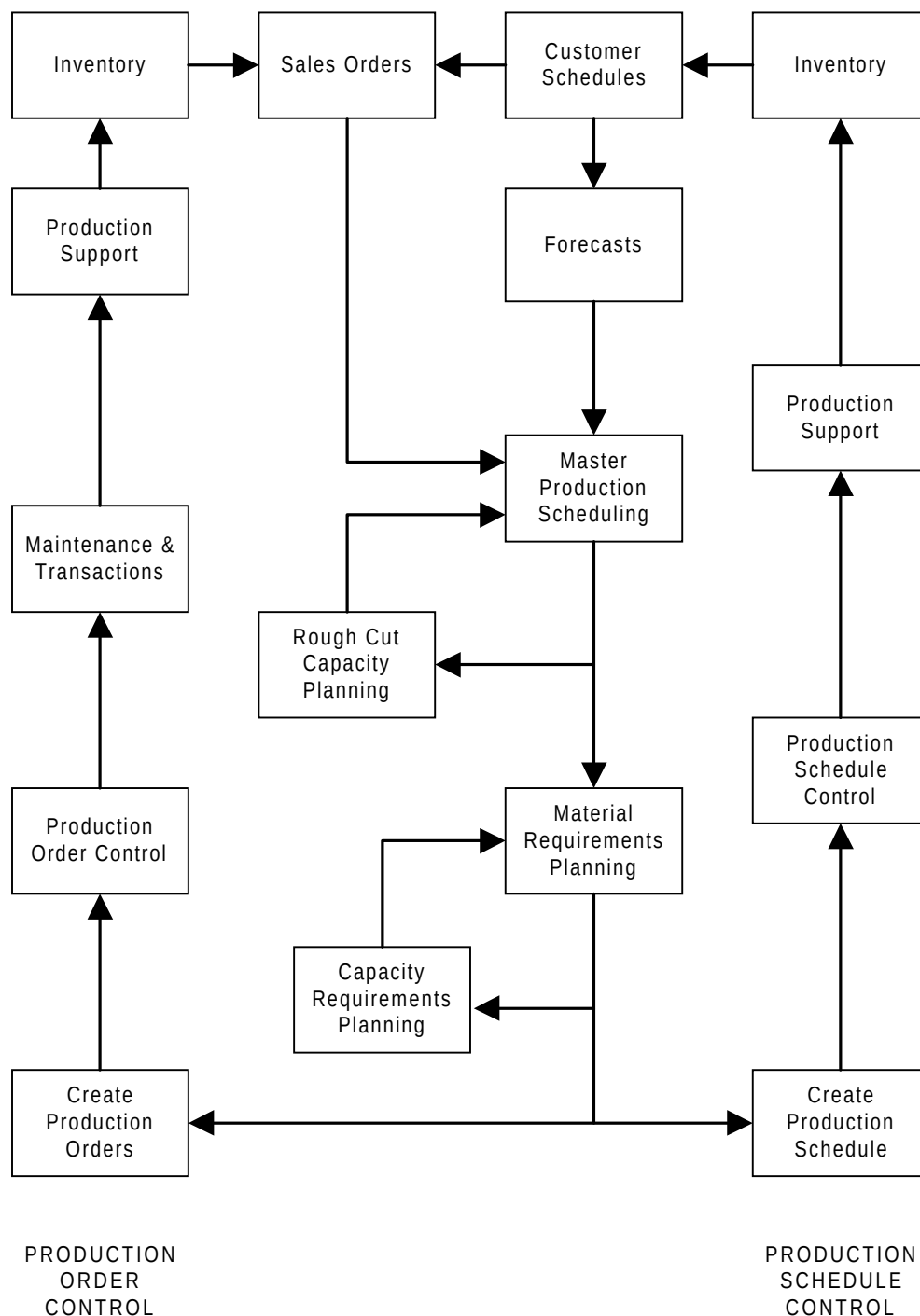
The capacity of the facility to implement the high level plan may then be checked using the Rough Cut Capacity Planning facility and if necessary the MPS planning cycle re-run.

Once a plan is agreed, the suggested orders output from the MPS cycle are then combined with associated component items and raw materials to create a demand input to the Material Requirements Planning (MRP) application. MRP reconciles existing production commitments with the revised MPS demand to create a more detailed production and purchase order plan. This plan suggests all purchase and production orders relevant to a user-defined planning horizon.

Again, before confirming any aspect of the new plan, the capacity of the Production facility may be checked using the full Capacity Requirements Planning (CRP) application to ensure that with realistic loading of the facility, it is possible to achieve the plan. This may require a number of iterations of the MRP planning cycle. Rough cut is not used at this point as it is assumed all planning details have been finalised.

When the production plan is finally approved, production and purchase orders are created and the planning cycle passes to the Production cycle.

Figure 5 - The Operation Management Cycle



The Production cycle enables orders of varying status to be released to production. Once released, orders may be tracked by notifying the system of operations and transactions completed through the Maintenance & Transactions functions. Production Support enables the entry of transactions and interrogation of the system through enquiry screens. Production reports may also be created. Completed items pass into Inventory to complete the Operations cycle.

Production Schedule Control

Production schedules are created by the Production Planning applications for specially designated schedule controlled items. These plan the availability of operations necessary to achieve a pre-determined volume of output, for a particular item using the available production facility.

The basic process is the same as for production orders. Customer schedules may be used to provide forecasts which drive the MPS and MRP applications as described for Production Order Control. The plans generated are suggested schedules and suggested purchase orders.

The output is a work station Production schedule. The Production cycle with respect to scheduling is slightly less complex as it considers only the end item or product, not the individual production orders as does Production Order Control.

Production Definition Management

Inventory is the source of the primary item database within the overall system environment for both purchased and production items. Every item to be referenced within Production must first be defined to Inventory.

Production Definition Management (PDM), Figure 6, manages the Production database which contains extended, Production specific details of all items used by the Production applications; definitions of the resources that used to combine raw material into finished products; and provides facilities to cost products.

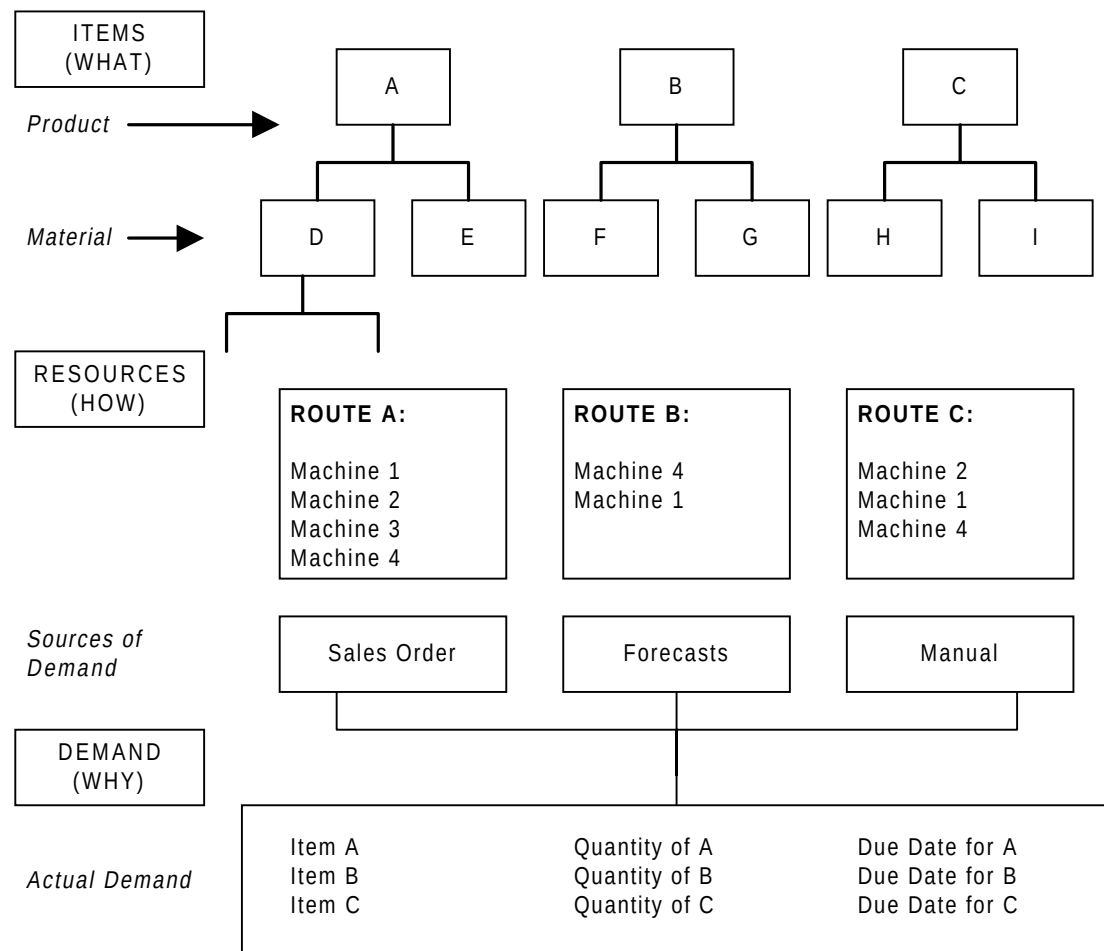
Resources are classified as work stations (which may be grouped to form work centres), the cost basis of each work station being defined by an assigned cost centre.

Single level production routes which are entered within the database define the operations required to make an item; the work stations at which the work is carried out; and materials necessary to complete the operation. The end result, or output, of a production route may be a single item such as a car, component or machine tool; or multiple outputs such as conserves and jams. Both single and multiple outputs may also produce waste and/or by-products. The single level production routes are automatically extended into full multi level structures by the system.

Database configuration in conjunction with the default settings in the Company Profile are fundamental to an effective business management system. A clear understanding of the database, its links to Inventory and the Company Profile is pre-requisite to determining the system to be implemented.

PDM is available in two forms, which are known as Base and Extended. The Base form supports all functionality available in previous releases of the software. The Extended form supports additional functionality, specifically schedule driven repetitive production; process production with multi product outputs and batch potency facilities.

Figure 6 - Production Definition Management



Master Production Scheduling

The Master Production Scheduling (MPS) application, Figure 7, produces a high level production plan of the critical and sales items. The system then matches the projected demand for designated products to the ability to supply based on user-defined planning models and planning horizons.

Demand is a function of sales orders, forecast, and/or manual entries or a user-defined combination. The ability to supply is a function of the availability of associated items and resources held in the Production database. The resulting plan may be based on production orders, production schedules or a mixture of both. The calculation process for the respective plan generations is simply:

Demand less Availability=Requirement

A demand forecast may be determined in a number of ways:

- Using a simple annual forecast defined within MPS.
- Using the output from the Forecasting application which takes account of historical sales analysis.
- Using the Customer Schedules application. This demand is raised remotely by customers at their facility and communicated across EDI links as part of an integrated supply chain.

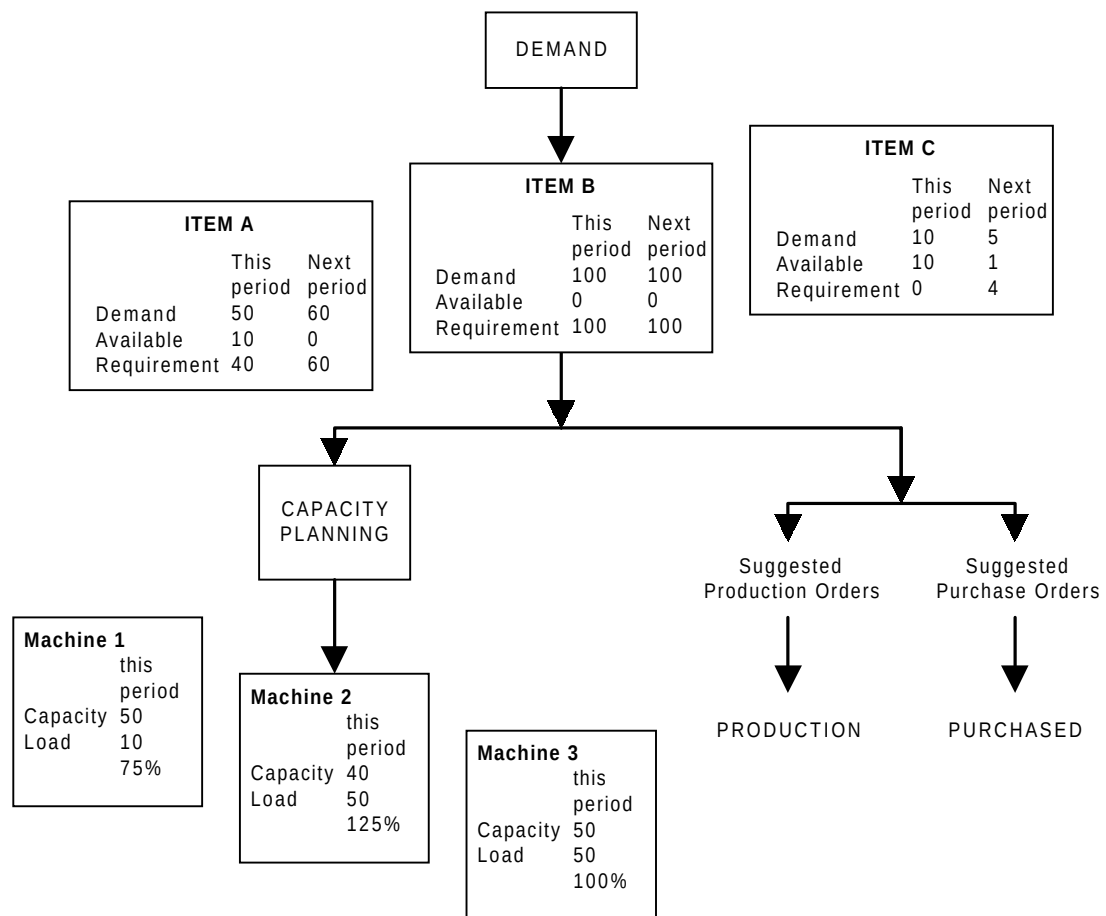
MPS is a time based plan of suggestions to meet customer due dates. The system works backwards from the customer due date specified, taking account of product lead times held within the Production database, to suggest a latest start date that achieves completion by the due date.

In each case, a plan is derived that also lists suggested purchase orders and Production resource requirements across the chosen time horizon. These may be individually confirmed and purchase orders automatically passed through to the Purchase Management application in advance of further MRP planning, if required.

Tactical line loading allows the re-scheduling of special flow routes for schedule controlled items; ensuring that all operations in an item schedule are replanned together regardless of the operation selected.

MPS also incorporates rough cut capacity planning. This may use summary routings to determine the loading in hours on the Production facility resources relative to their capacity in hours as defined in the production database.

Figure 7 - Master Production Scheduling



Material Requirements Planning

Material Requirements Planning (MRP), Figure 8, produces a more detailed plan than MPS, but is essentially completing the same function more comprehensively. MRP takes account of all the lower level items, sub-parents and raw materials. Planning is again based upon user-defined models.

MRP differs from MPS in two key respects. Firstly, it includes all items in a structure. Secondly, there is a choice of three modes of processing:

- Regenerative: Plan for all items across all structures.
- Selective: Plan only selected items and hence selected structures.
- Net Change: Plan only items that have created a net change 'trigger' as a result of selected transactions or database changes.

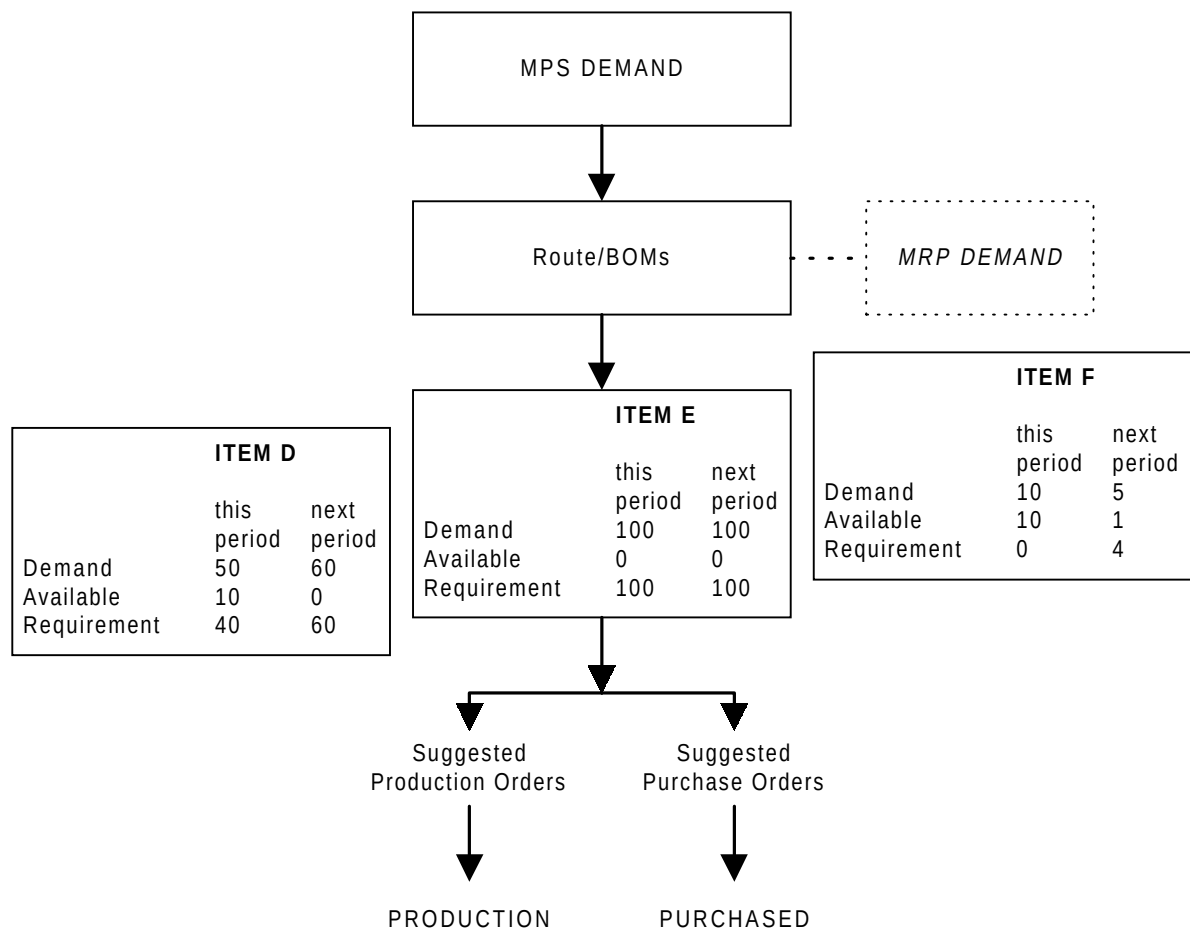
A cell based view of the above is also possible. A cell based MRP run determines supply and demand details for cells (departments or sub-divisions of a Company/Organisational Model).

MRP and MPS are autonomous modules. Either may be used stand alone. However, MRP is normally driven by MPS demand as shown in Figure 8. A separate MRP demand may be used to enhance the requirements for dependant MRP items.

MRP and MPS plans are based upon independent user-defined models and reporting profiles. The reporting profiles may be based on a non-linear time horizon, for example, a three month plan might be structured as follows:

Next Week	Daily detail
Week 2-4	Weekly detail
Week 5-8	Fortnightly detail
Week 8-12	Monthly detail

Figure 8 - Material Requirements Planning



Capacity Requirements Planning

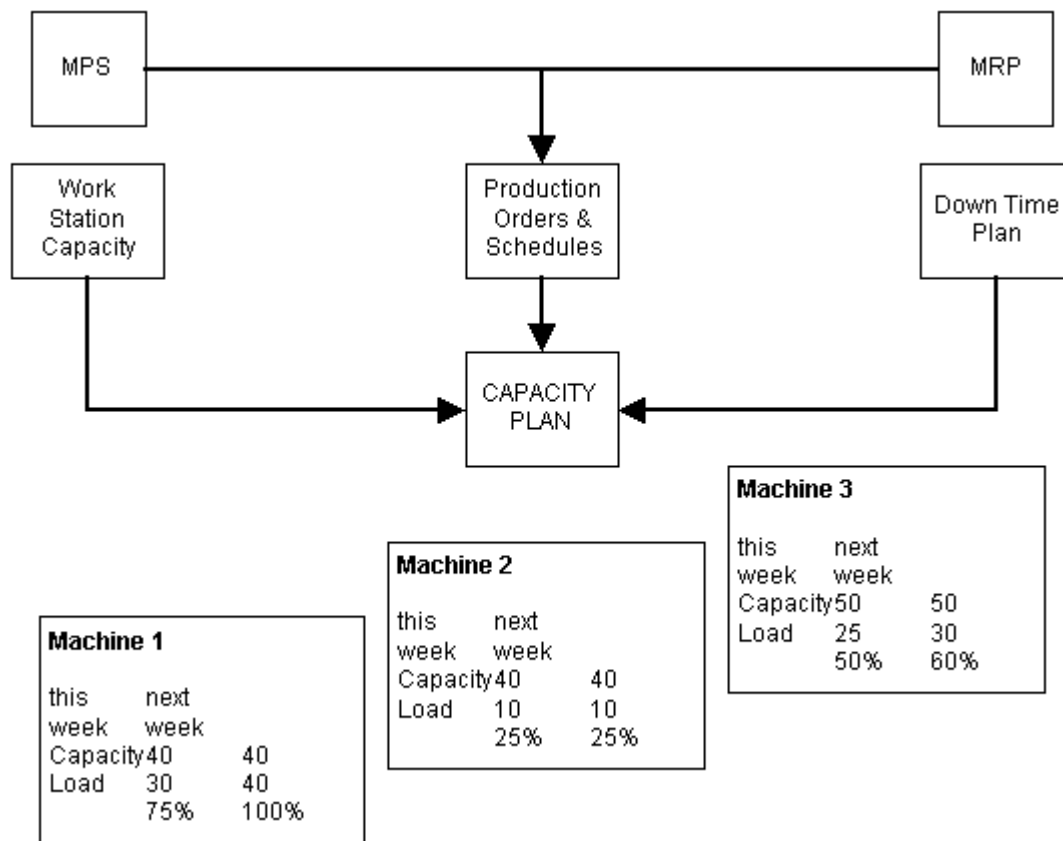
Capacity Requirements Planning, Figure 9, takes account of work station capacities, and planned down times defined across the whole Production application. It uses this information to assess the demand proposed by MPS or MRP plans and determines a loading factor for each work station.

Database item/route information is accessed to convert MPS/MRP supply requirements (in quantities) into the number of hours required (the load) at each work station. MPS/MRP planned supply dates are used to schedule these required hours into appropriate weekly production time slots at each work station, with reference to work station standard capacities and with due regard to planned down times and non working days affecting the planning run timescales.

Reports and enquiries are available which compare the weekly work station/centre hours required and available within the planning run timescales, so that it is possible to determine whether over/underload capacity situations occur and thus to decide on action(s) required.

The application enables finite capacity planning by individual order and schedule, but assumes infinite capacity of the facility. It is therefore the responsibility of the planner to complete any fine-tuning of the work load to optimise the loading factor and hence utilisation of production capacity.

Figure 9 - Capacity Requirements Planning



Production Control

Production Control, Figure 10, is the combined functionality of the Production Order Control and Production Schedule Control and Production Support applications. These provide the means to record the 'real world' implementation of the production plan. Production Order Control and Production Schedule Control permit operational transactions to be made against the plan.

Production Order Control

Production Order Control enables the Production system to monitor the progress of orders through the Production facility by logging the status of each order within a work station. The order status may be:

- Suggested.
- Planned.
- Confirmed.
- Released.
- Completed.
- Cancelled.

In addition, Production Order Control facilitates the recording of operational transactions; the issue of materials, backflushing of bulk issues, scrap, re-work and placing unused items back in stock.

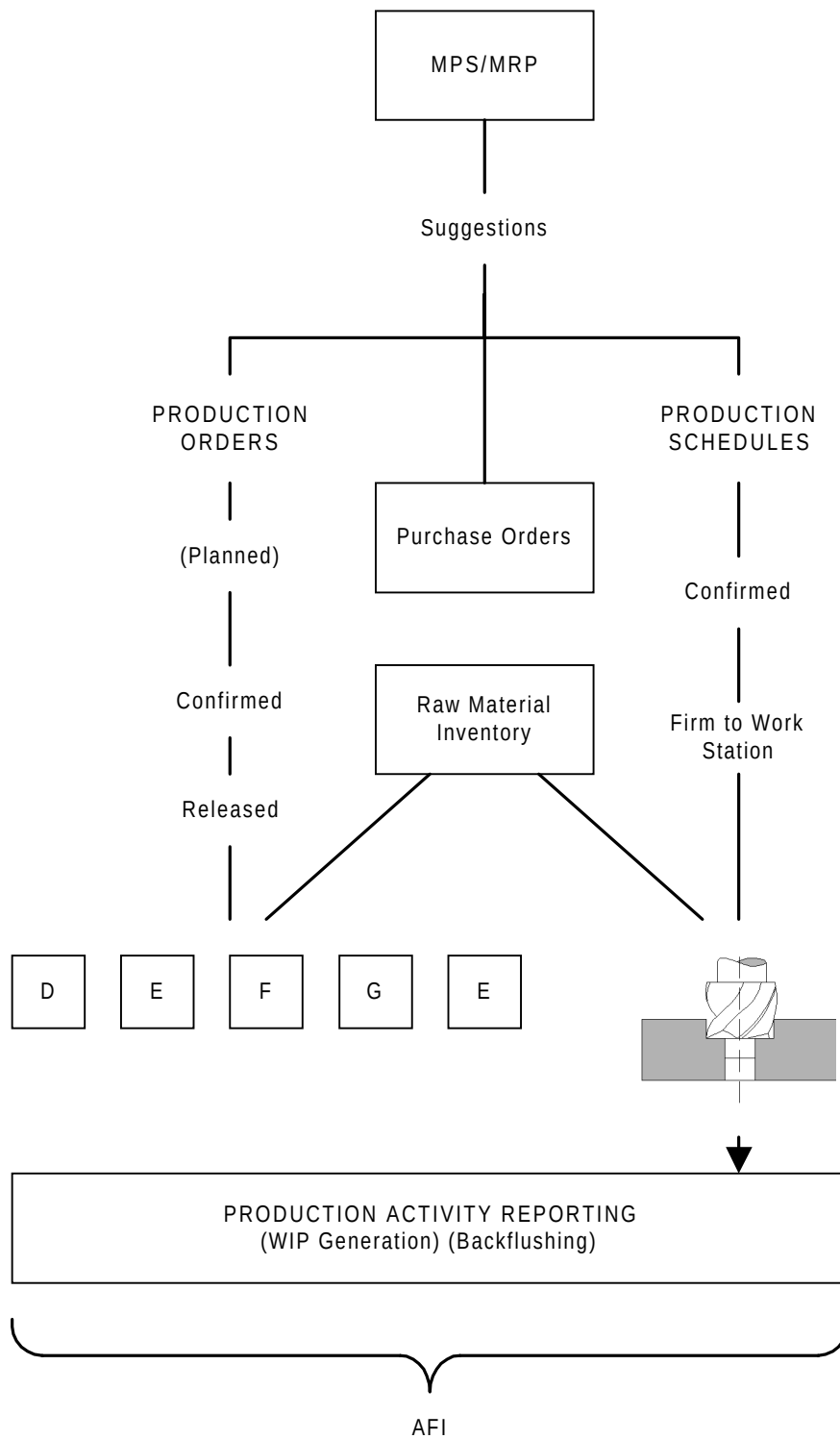
Production Schedule Control

Production Schedule Control provides similar facilities to Production Order Control, but for items subject to schedule control.

Production Support

Production Support facilities assist in the management of work stations, work-in-progress (WIP) and Inventory stocks. Production related reports may be generated which show costing and efficiency information. The accuracy of these reports is dependant on maintaining good discipline in booking operations. The rigor of such discipline has to be balanced with the demands of the Production environment.

Figure 10 - Production Control



Advanced Financial Integrator (AFI)

A link is provided to the General Ledger through the AFI application. AFI makes use of the transactions booked to the Production application enabling the calculation of financial information.